

Advanced Practical AI

Five connected sessions. One complete reliable AI system.

**DATA -> GENERALIZATION -> EXPLANATION -> RELIABILITY ->
PRODUCT**

SESSION OBJECTIVE

Build one AI system from defective raw sources to a tested decision application.

5 connected sessions

5 practical artifacts

5 assignments

1 final application

PRIMARY ARTIFACT

Reliable AI Risk Intelligence

THE CORE QUESTION

Can an AI system be trusted when the score looks impressive?

The course answers this with evidence across data, validation, explanations, reliability, and productization.

THE COMPLETE SYSTEM



Every session adds one layer. Nothing is treated as an isolated notebook exercise.

FIVE SESSIONS - FIVE ARTIFACTS

01 DATA FUSION

AI Data Fusion Inspector

02 VALIDATION

Model Selection and Leakage Report

03 EXPLAINABILITY

AI Explanation Casebook

04 RELIABILITY

Reliable AI Decision Router

05 PRODUCTIONIZATION

Working Risk Intelligence App

LIVE WORKSPACE RHYTHM

OBJECTIVE

One measurable destination.

PREDICT

Record an expectation before execution.

GUIDED BUILD

Important logic stays visible.

BREAK IT

One deliberate failure is created.

FIX IT

The failure is corrected with evidence.

CHECKPOINT

Assertions verify the contract.

MINI-PROJECT

A meaningful artifact is exported.

EVIDENCE

A decision record is produced.

ASSIGNMENT

Independent work is isolated from solutions.

WHY TWO NOTEBOOK SETS

LIVE WORKSPACES

- TODO cells remain visible
- Prediction checkpoints
- Failure is experienced
- Reasoning precedes abstraction

COMPLETE REFERENCE

- Every cell implemented
- Heavy explanatory comments
- Expected assertions included
- Reusable after the live session

The live path protects effort. The complete path protects long-term reuse.

THE MODEL IS ONE COMPONENT



A strong algorithm cannot repair a broken observation grain, leaked future information, or unsafe routing.

BUILT-IN EVIDENCE

840

trusted rows

one region x one month

0

duplicate keys

after cleaning

16

model features

raw + engineered

2

prohibited fields

retained only for leakage demo

The deterministic data package creates reproducible checkpoints while preserving noisy, overlapping classes.

MODEL COMPARISON USED IN SESSION 2

Candidate	Macro F1	Eligibility	Decision
Dummy	0.2016	Legitimate	Baseline
Logistic	0.8008	Legitimate	Selected
XGBoost corrected	0.7466	Legitimate	Not selected
XGBoost leaky	0.9150	Invalid	Excluded before ranking

The selection rule follows evidence: highest legitimate macro F1, then lowest log loss.

A DELIBERATE SURPRISE

The complex model does not win.

The simpler logistic pipeline is persisted because it performs better on the legitimate future test.
Complexity receives no automatic reward.

RELIABLE AI ROUTING

ACCEPT

Input is valid, pattern is not anomalous, and confidence meets policy.

HUMAN REVIEW

Anomaly or low-confidence evidence requires review.

REJECT

The external input contract is violated before prediction.

SOURCE PACKAGE



Everything required for delivery, independent effort, complete solutions, and final execution is separated by purpose.

EVALUATION EVIDENCE

1

Data contract: exact row grain, duplicate keys, missing model cells, and prohibited features.

2

Generalization: chronological split with 2025 reserved as unseen future test data.

3

Model choice: leaky candidate excluded before metric ranking.

4

Explanation: global permutation importance, local occlusion, and optimized Tree SHAP for the tree case.

Reliability: calibration analysis, selective prediction, anomaly review, and hard input rejection.

The final demonstration follows the same evidence chain.

6

Productization: exact selected model is persisted and verified by automated tests.

FINAL DESTINATION

A reproducible AI decision system that can explain what it did, refuse invalid requests, defer uncertain cases, and rebuild from source.